Data Mining Applications

- Data Mining for Sentiment and Opinion
- Truth Discovery and Misinformation Identification
- Information and Disease Propagation
- Productivity and Team Science

Information and Disease Propagation

- Data mining research problems:
  - Prediction problems of propagation.
  - Optimization problems of propagation.
- Applications:
  - Social media: predict which piece of information is likely to go viral; detect the rumor source who started a misinformation campaign; neutralize the propagation of misinformation.
  - Computational epidemiology: determine critical network condition (e.g., epidemic threshold) under which an epidemic is likely to happen.

Information and Disease Propagation

- Prediction problems of propagation:
  - Classification and regression: whether information being popular/popular scores.
  - Prediction around publication: before/after information is published.
  - Prediction at different granularities: individual/groups of information.
- Optimization problems of propagation (in a social network perspective):
  - Influence maximization: choose a set of initial nodes to maximize the number of infected nodes.
  - Source localization: identify source of a cascade over network with partial observations.
  - Activity shaping: steer user's activity.
  - Graph connectivity optimization: manipulate the graph topology to affect the information propagation results.

Information and Disease Propagation

- Models for information diffusion problems:
  - Feature based model:
    - Idea: utilize supervised models for information diffusion prediction.
    - Features used: temporal, local and global structure, user and item, etc.
  - Generative model:
    - Idea: take information diffusion as event sequences in the continuous temporal domain, and formulate as statistical generative approaches.
    - Methods: Poisson Process, Survival Analysis, Hawkes Process, Epidemic Model.
  - Deep learning based model:
    - Idea: no assumptions on the information diffusion process.
    - Methods: multi-modal data with various neural modules.
- Computational epidemiology models: SIS, SIR and SEIR and their variants.